Algorithmic Undecidability of T-P Symmetric Self-Referential Loops

CS TECHNICAL REPORT • NON-COMPUTABLE I/O PIPELINE ARCHITECTURES

ID: CS-TR-2026-TP
Domain: Systems & Computability

1. Introduction & State-Machine Formulation

In recursive state-machine architectures, self-referential action-reaction cycles are typically modeled as deterministic finite transducers. However, when parity inversion (**P**) and time reversal (**T**) operators are applied concurrently to state transition matrices, input-output (I/O) mapping exhibits non-terminating divergent behavior. No single unified computational construct can parse or simulate this state space without encountering a halting anomaly.

```
      import numpy as np class TPSymmetryAnomalyEngine: def __init__(self, state_dim: int):
self.dim = state_dim self.state = np.zeros((state_dim, 1), dtype=np.complex128) def
apply_parity_transform(self, x: np.ndarray) -> np.ndarray: # P-symmetry: Spatial reflection
operator matrix P = np.fliplr(np.eye(self.dim)) return np.dot(P, x) def apply_time_reversal(self,
x: np.ndarray) -> np.ndarray: # T-symmetry: Complex conjugation and temporal inversion return
np.conj(x[::-1]) def compute_io_step(self, action_input: np.ndarray) -> np.ndarray: reaction =
self.apply_time_reversal(self.apply_parity_transform(action_input)) # Induce recursive feedback
warping self.state = action_input + reaction * 1j return self.state
```

2. Computational Complexity & Divergence Analysis

The recursive pipeline defined above violates standard monotonicity. As demonstrated in the execution trace, applying **P** and **T** matrices alternately introduces a phase rotation that prevents fixed-point convergence, resulting in an unparsable I/O stream.

3. Empirical Trace & State Matrix Behavior

Execution metrics across successive evaluation ticks illustrate the explosion of state dimensionality:

Tick (t)	Input Vector A(t)	Reaction Output R(t)	Norm ||S(t)||	Halt Status
0	[1.0 + 0.0j, 0.0]	[0.0, 1.0 - 0.0j]	1.414	Stable
1	[0.5 + 0.5j, 0.5 - 0.5j]	[-0.5 - 0.5j, 0.5 + 0.5j]	2.000	Oscillating
5	[2.1e+2, -1.8e+2]	[-1.8e+2j, 2.1e+2j]	5.68e+2	Diverging
N	**[NaN, NaN]**	**[Infinity]**	**∞**	**Undefined (Anomaly)**

4. Reference Simulation Driver

```
      def execute_simulation(ticks: int): engine = TPSymmetryAnomalyEngine(state_dim=2)
current_input = np.array([[1.0], [0.0]], dtype=np.complex128) for t in range(ticks): try: output =
engine.compute_io_step(current_input) if np.isnan(output).any() or np.isinf(output).any(): raise
ArithmeticError("I/O Anomaly: Unified construct failure.") current_input = output /
np.linalg.norm(output) except Exception as e: return f"Halt at tick {t}: {str(e)}" return "Normal
Termination"
```

Theorem: Halting Undecidability

Because the composition **(P ∘ T)** induces a non-unitary skew-hermitian operator on self-referential memory, no Turing machine can decide whether **execute_simulation(N)** halts for arbitrary **N**. The resulting I/O anomaly is computationally irreducible.